CSE 3009-No SQL
Slot:L7+L8
Bus Ticket Booking using MongoDB

Guided By:

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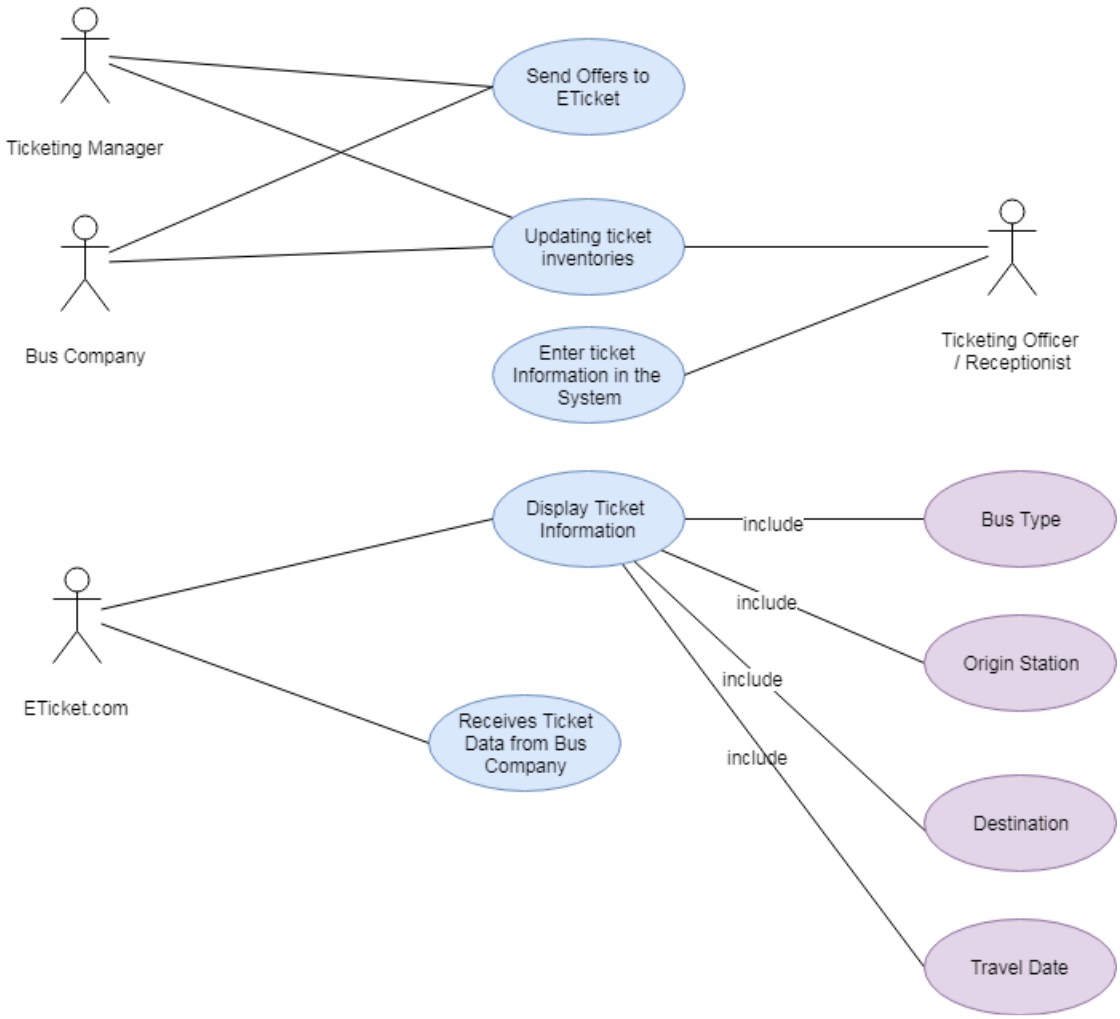
Use Case Diagram



Queries

1.
```
> show dbs
admin   0.000GB
config  0.000GB
local   0.000GB
mydb    0.000GB
>
```

2.
```
> use data
switched to db data
>
```

3.
```
> db
data
```

4.
```
> db.createCollection("bus")
{ "ok" : 1 }
>
```

5.
```
> db.bus.insert({bus_id:142,name:"sravani",start_place:"Rajahmundry",destination:"Visakhapatnam"
,ticket_cost:2000})
WriteResult({ "nInserted" : 1 })
>
```

6.
```
> db.bus.insertMany([{bus_id:200,name:"meghana",start_place:"Narasaraopeta",destination:"Banglore",ticket_cost:5000},{bu
s_id:350,name:"harsha",start_place:"Guntur",destination:"Munnar",ticket_cost:8000}])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("6230b062fbd872e0c3bc86d3"),
                ObjectId("6230b062fbd872e0c3bc86d4")
        ]
}
```

7.
```
> db.bus.find()
{ "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230af00fbd872e0c3bc86d2"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 2500 }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d3"), "bus_id" : 200, "name" : "meghana", "start_place" : "Narasaraopeta", "de
stination" : "Banglore", "ticket_cost" : 5000 }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d4"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 8000 }
>
```

8.
```
> db.bus.find().skip(1)
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 2500 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Narasaraopeta", "de
stination" : "Banglore", "ticket_cost" : 5000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 8000 }
>
```

9.

```
> db.bus.find().limit(2)
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 2500 }
>
```

10.

```
> db.bus.find().pretty()
{
        "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230af00fbd872e0c3bc86d2"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
{
        "_id" : ObjectId("6230b062fbd872e0c3bc86d3"),
        "bus_id" : 200,
        "name" : "meghana",
        "start_place" : "Narasaraopeta",
        "destination" : "Banglore",
        "ticket_cost" : 5000
}
{
        "_id" : ObjectId("6230b062fbd872e0c3bc86d4"),
        "bus_id" : 350,
        "name" : "harsha",
        "start_place" : "Guntur",
        "destination" : "Munnar",
        "ticket_cost" : 8000
}
>
```

11.
```
> db.bus.find().pretty().limit(2)
{
        "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
>
```

12.
```
> db.bus.find().pretty().skip(2)
{
        "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"),
        "bus_id" : 200,
        "name" : "meghana",
        "start_place" : "Narasaraopeta",
        "destination" : "Banglore",
        "ticket_cost" : 5000
}
{
        "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"),
        "bus_id" : 350,
        "name" : "harsha",
        "start_place" : "Guntur",
        "destination" : "Munnar",
        "ticket_cost" : 8000
}
>
```

13.

```
> db.bus.find().forEach(printjson)
{
        "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230af00fbd872e0c3bc86d2"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
{
        "_id" : ObjectId("6230b062fbd872e0c3bc86d3"),
        "bus_id" : 200,
        "name" : "meghana",
        "start_place" : "Narasaraopeta",
        "destination" : "Banglore",
        "ticket_cost" : 5000
}
{
        "_id" : ObjectId("6230b062fbd872e0c3bc86d4"),
        "bus_id" : 350,
        "name" : "harsha",
        "start_place" : "Guntur",
        "destination" : "Munnar",
        "ticket_cost" : 8000
}
>
```

14.

```
> db.bus.find().limit(2).forEach(printjson)
{
        "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
>
```

15.
```
> db.bus.find().skip(2).forEach(printjson)
{
        "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"),
        "bus_id" : 200,
        "name" : "meghana",
        "start_place" : "Narasaraopeta",
        "destination" : "Banglore",
        "ticket_cost" : 5000
}
{
        "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"),
        "bus_id" : 350,
        "name" : "harsha",
        "start_place" : "Guntur",
        "destination" : "Munnar",
        "ticket_cost" : 8000
}
>
```

16.
```
> db.bus.createIndex({"bus_id":1})
{
        "numIndexesBefore" : 1,
        "numIndexesAfter" : 2,
        "createdCollectionAutomatically" : false,
        "ok" : 1
}
>
```

17.
```
> db.bus.getIndexes()
[
        {
                "v" : 2,
                "key" : {
                        "_id" : 1
                },
                "name" : "_id_"
        },
        {
                "v" : 2,
                "key" : {
                        "bus_id" : 1
                },
                "name" : "bus_id_1"
        }
]
>
```

18.
```
> db.bus.dropIndexes()
{
        "nIndexesWas" : 2,
        "msg" : "non-_id indexes dropped for collection",
        "ok" : 1
}
>
```

19.
```
> db.bus.createIndex({"ticket_cost":1})
{
        "numIndexesBefore" : 1,
        "numIndexesAfter" : 2,
        "createdCollectionAutomatically" : false,
        "ok" : 1
}
>
```

20.
```
> db.bus.getIndexes()
[
        {
                "v" : 2,
                "key" : {
                        "_id" : 1
                },
                "name" : "_id_"
        },
        {
                "v" : 2,
                "key" : {
                        "ticket_cost" : 1
                },
                "name" : "ticket_cost_1"
        }
]
>
```

21.
```
> db.bus.dropIndexes()
{
        "nIndexesWas" : 2,
        "msg" : "non-_id indexes dropped for collection",
        "ok" : 1
}
>
```

22.
```
> db.bus.insertMany([{bus_id:210,name:"satya",start_place:"Banglore",destination:"Delhi",ticket_cost:5000},{bus_id:415,n
ame:"vahini",start_place:"Yanam",destination:"Andaman",ticket_cost:2500}])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("6230b31ffbd872e0c3bc86d5"),
                ObjectId("6230b31ffbd872e0c3bc86d6")
        ]
}
>
```

23.
```
> db.bus.find({"ticket_cost":{$lt:4000}})
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 2500 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 2500 }
>
```

24.

```
> db.bus.find({"ticket_cost":{$lt:4000}}).pretty()
{
        "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
{
        "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"),
        "bus_id" : 415,
        "name" : "vahini",
        "start_place" : "Yanam",
        "destination" : "Andaman",
        "ticket_cost" : 2500
}
>
```

25.

```
> db.bus.find({"ticket_cost":{$lt:5000}}).limit(2).pretty()
{
        "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
>
```

26.
```
> db.bus.find({"ticket_cost":{$lt:5000}}).limit(2).skip(1).pretty()
{
        "_id" : ObjectId("6230af00fbd872e0c3bc86d2"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
{
        "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"),
        "bus_id" : 415,
        "name" : "vahini",
        "start_place" : "Yanam",
        "destination" : "Andaman",
        "ticket_cost" : 2500
}
>
```

27.
```
> db.bus.find({},{"start_place":1,"bus_id":1}).sort({ticket_cost:1})
{ "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"), "bus_id" : 142, "start_place" : "Rajahmundry" }
{ "_id" : ObjectId("6230af00fbd872e0c3bc86d2"), "bus_id" : 150, "start_place" : "Nuzvid" }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"), "bus_id" : 415, "start_place" : "Yanam" }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d3"), "bus_id" : 200, "start_place" : "Narasaraopeta" }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d5"), "bus_id" : 210, "start_place" : "Banglore" }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d4"), "bus_id" : 350, "start_place" : "Guntur" }
>
```

28.
```
> db.bus.find({},{"start_place":1,"bus_id":1}).sort({bus_id:-1})
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "start_place" : "Yanam" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "start_place" : "Guntur" }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "start_place" : "Banglore" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "start_place" : "Narasaraopeta" }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "start_place" : "Nuzvid" }
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "start_place" : "Rajahmundry" }
>
```

29.
```
> db.bus.find({"ticket_cost":{$gte:2500}})
{ "_id" : ObjectId("6230af00fbd872e0c3bc86d2"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 2500 }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d3"), "bus_id" : 200, "name" : "meghana", "start_place" : "Narasaraopeta", "de
stination" : "Banglore", "ticket_cost" : 5000 }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d4"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 8000 }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d5"), "bus_id" : 210, "name" : "satya", "start_place" : "Banglore", "destinati
on" : "Delhi", "ticket_cost" : 5000 }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 2500 }
>
```

30.
```
> db.bus.find({"bus_id":{$gte:210}})
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 8000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Banglore", "destinati
on" : "Delhi", "ticket_cost" : 5000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 2500 }
>
```

31.
```
> db.bus.find({"ticket_cost":{$lt:5000}}).pretty()
{
        "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230af00fbd872e0c3bc86d2"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
{
        "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"),
        "bus_id" : 415,
        "name" : "vahini",
        "start_place" : "Yanam",
        "destination" : "Andaman",
        "ticket_cost" : 2500
}
>
```

32.
```
> db.bus.find({"bus_id":{$lt:210}}).pretty()
{
        "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
{
        "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"),
        "bus_id" : 200,
        "name" : "meghana",
        "start_place" : "Narasaraopeta",
        "destination" : "Banglore",
        "ticket_cost" : 5000
}
>
```

33.

```
> db.bus.find({"name":{$ne:"sravani"}}).pretty()
{
        "_id" : ObjectId("6230af00fbd872e0c3bc86d2"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
{
        "_id" : ObjectId("6230b062fbd872e0c3bc86d3"),
        "bus_id" : 200,
        "name" : "meghana",
        "start_place" : "Narasaraopeta",
        "destination" : "Banglore",
        "ticket_cost" : 5000
}
{
        "_id" : ObjectId("6230b062fbd872e0c3bc86d4"),
        "bus_id" : 350,
        "name" : "harsha",
        "start_place" : "Guntur",
        "destination" : "Munnar",
        "ticket_cost" : 8000
}
{
        "_id" : ObjectId("6230b31ffbd872e0c3bc86d5"),
        "bus_id" : 210,
        "name" : "satya",
        "start_place" : "Banglore",
        "destination" : "Delhi",
        "ticket_cost" : 5000
}
{
        "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"),
        "bus_id" : 415,
        "name" : "vahini",
        "start_place" : "Yanam",
        "destination" : "Andaman",
        "ticket_cost" : 2500
}
>
```

34.
```
> db.bus.find({"start_place":{$in:["Guntur","Munnar","Banglore"]}}).pretty()
{
        "_id" : ObjectId("6230b062fbd872e0c3bc86d4"),
        "bus_id" : 350,
        "name" : "harsha",
        "start_place" : "Guntur",
        "destination" : "Munnar",
        "ticket_cost" : 8000
}
{
        "_id" : ObjectId("6230b31ffbd872e0c3bc86d5"),
        "bus_id" : 210,
        "name" : "satya",
        "start_place" : "Banglore",
        "destination" : "Delhi",
        "ticket_cost" : 5000
}
>
```

35.
```
> db.bus.find({"start_place":{$nin:["Yanam","Guntur"]}}).pretty()
{
        "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230af00fbd872e0c3bc86d2"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 2500
}
{
        "_id" : ObjectId("6230b062fbd872e0c3bc86d3"),
        "bus_id" : 200,
        "name" : "meghana",
        "start_place" : "Narasaraopeta",
        "destination" : "Banglore",
        "ticket_cost" : 5000
}
{
        "_id" : ObjectId("6230b31ffbd872e0c3bc86d5"),
        "bus_id" : 210,
        "name" : "satya",
        "start_place" : "Banglore",
        "destination" : "Delhi",
        "ticket_cost" : 5000
}
>
```

36.
```
> db.bus.update({bus_id:210},{$set:{"start_place":"Hyderabad"}})
WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1 })
>
```

37.
```
> db.bus.update({name:"meghana"},{$set:{"start_place":"Guntur"}})
WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1 })
>
```

38.
```
> db.bus.update({bus_id:150},{$set:{"Destination":"Araku"}})
WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1 })
>
```

39.
```
> db.bus.update({"name":"harsha"},{$set:{"ticket_cost":4000}})
WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1 })
>
```

40.
```
> db.bus.updateMany({"ticket_cost":{$gt:2000}},{$set:{"ticket_cost":4000}})
{ "acknowledged" : true, "matchedCount" : 5, "modifiedCount" : 4 }
>
```

41.
```
> db.bus.find()
{ "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230af00fbd872e0c3bc86d2"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d3"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d4"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d5"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
>
```

42.
```
> db.bus.updateMany({"ticket_cost":{$lt:4000}},{$set:{"ticket_cost":5000}})
{ "acknowledged" : true, "matchedCount" : 1, "modifiedCount" : 1 }
> db.bus.find()
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 5000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
>
```

43.
```
> db.bus.updateMany({"ticket_cost":{$gte:4100}},{$set:{"ticket_cost":2000}})
{ "acknowledged" : true, "matchedCount" : 1, "modifiedCount" : 1 }
> db.bus.find()
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
>
```

44.
```
> db.bus.find({$and:[{bus_id:142},{destination:"Visakhapatnam"}]}).pretty()
{
        "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
>
```

45.
```
> db.bus.find({$or:[{bus_id:150},{destination:"Munnar"}]}).pretty()
{
        "_id" : ObjectId("6230af00fbd872e0c3bc86d2"),
        "bus_id" : 150,
        "name" : "roopa",
        "start_place" : "Nuzvid",
        "destination" : "Hyderabad",
        "ticket_cost" : 4000,
        "Destination" : "Araku"
}
{
        "_id" : ObjectId("6230b062fbd872e0c3bc86d4"),
        "bus_id" : 350,
        "name" : "harsha",
        "start_place" : "Guntur",
        "destination" : "Munnar",
        "ticket_cost" : 4000
}
```

46.
```
> db.bus.find({$or:[{destination:"Munnar"},{$and:[{bus_id:142},{destination:"Visakhapatnam"}]}]}).pretty()
{
        "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"),
        "bus_id" : 350,
        "name" : "harsha",
        "start_place" : "Guntur",
        "destination" : "Munnar",
        "ticket_cost" : 8000
}
>
```

47.
```
> db.bus.find({$and:[{destination:"Munnar"},{$or:[{bus_id:350},{destination:"Visakhapatnam"}]}]}).pretty()
{
        "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"),
        "bus_id" : 350,
        "name" : "harsha",
        "start_place" : "Guntur",
        "destination" : "Munnar",
        "ticket_cost" : 8000
}
>
```

48.
```
> db.bus.find({ticket_cost:{$not:{$gt:2100}}}).pretty()
{
        "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
>
```

49.
```
> db.bus.find({$or:[{destination:"Munnar"},{ticket_cost:{$not:{$gt:2100}}}]}).pretty()
{
        "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"),
        "bus_id" : 142,
        "name" : "sravani",
        "start_place" : "Rajahmundry",
        "destination" : "Visakhapatnam",
        "ticket_cost" : 2000
}
{
        "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"),
        "bus_id" : 350,
        "name" : "harsha",
        "start_place" : "Guntur",
        "destination" : "Munnar",
        "ticket_cost" : 8000
}
>
```

50.
```
> db.bus.find({bus_id:{$gt:400}},{"start_place":1,"bus_id":1}).sort({ticket_cost:1})
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "start_place" : "Yanam" }
>
```

51.
```
> db.bus.find().sort({name:-1})
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230e211fc8fd25c07c5c47b"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
>
```

52.
```
> db.bus.find({},{ticket_cost:1}).sort({ticket_cost:-1})
{ "_id" : ObjectId("6230af00fbd872e0c3bc86d2"), "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d3"), "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d4"), "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d5"), "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"), "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"), "ticket_cost" : 2000 }
>
```

53.
```
> db.bus.find({},{ticket_cost:1}).sort({ticket_cost:1})
{ "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"), "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230af00fbd872e0c3bc86d2"), "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d3"), "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d4"), "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d5"), "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"), "ticket_cost" : 4000 }
>
```

54.
```
> db.bus.find({},{bus_id:1}).sort({ticket_cost:1})
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415 }
```

55.
```
> db.bus.find({},{name:1}).sort({bus_id:-1})
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "name" : "vahini" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "name" : "harsha" }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "name" : "satya" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "name" : "meghana" }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "name" : "roopa" }
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "name" : "sravani" }
```

56.
```
> db.bus.insert({bus_id:453,name:"sita",start_place:"Vijayawada",destination:"Mumbai",ticket_cost:8000})
WriteResult({ "nInserted" : 1 })
>
```

57.
```
> db.bus.insertMany([{bus_id:521,name:"ganesh",start_place:"Ramachandrapuram",destination:"Rajahmundry",ticket_cost:500}
,{bus_id:111,start_place:"Guntur",destination:"Amaravati",ticket_cost:850}])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("62315edc45c23ff7cbde1923"),
                ObjectId("62315edc45c23ff7cbde1924")
        ]
}
>
```

58.
```
> db.bus.insertOne({bus_id:253,name:"geeta",start_place:"Hyderabad",destination:"Chennai",ticket_cost:5500})
{
        "acknowledged" : true,
        "insertedId" : ObjectId("62315f4145c23ff7cbde1925")
}
>
```

59.
```
> db.bus.find({}).skip(1).sort({name:-1})
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230e211fc8fd25c07c5c47b"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("62315df545c23ff7cbde1922"), "bus_id" : 453, "name" : "sita", "start_place" : "Vijayawada", "destinat
ion" : "Mumbai", "ticket_cost" : 8000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("62315f4145c23ff7cbde1925"), "bus_id" : 253, "name" : "geeta", "start_place" : "Hyderabad", "destinat
ion" : "Chennai", "ticket_cost" : 5500 }
{ "_id" : ObjectId("62315edc45c23ff7cbde1923"), "bus_id" : 521, "name" : "ganesh", "start_place" : "Ramachandrapuram", "
destination" : "Rajahmundry", "ticket_cost" : 500 }
{ "_id" : ObjectId("62315edc45c23ff7cbde1924"), "bus_id" : 111, "start_place" : "Guntur", "destination" : "Amaravati", "
ticket_cost" : 850 }
>
```

60.
```
> db.bus.deleteOne({"name":"sita"})
{ "acknowledged" : true, "deletedCount" : 1 }
>
```

61.
```
> db.bus.deleteOne({"ticket_cost":850})
{ "acknowledged" : true, "deletedCount" : 1 }
> db.bus.find()
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230e211fc8fd25c07c5c47b"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("62315edc45c23ff7cbde1923"), "bus_id" : 521, "name" : "ganesh", "start_place" : "Ramachandrapuram", "
destination" : "Rajahmundry", "ticket_cost" : 500 }
{ "_id" : ObjectId("62315f4145c23ff7cbde1925"), "bus_id" : 253, "name" : "geeta", "start_place" : "Hyderabad", "destinat
ion" : "Chennai", "ticket_cost" : 5500 }
>
```

62.
```
> db.bus.deleteMany({"ticket_cost":{$lt:1000}})
{ "acknowledged" : true, "deletedCount" : 1 }
> db.bus.find()
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230e211fc8fd25c07c5c47b"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("62315f4145c23ff7cbde1925"), "bus_id" : 253, "name" : "geeta", "start_place" : "Hyderabad", "destinat
ion" : "Chennai", "ticket_cost" : 5500 }
>
```

63.
```
> db.bus.deleteMany({"ticket_cost":5500})
{ "acknowledged" : true, "deletedCount" : 1 }
> db.bus.find()
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230e211fc8fd25c07c5c47b"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
>
```

64.
```
> db.bus.insert({bus_id:452,name:"sita",start_place:"Vijayawada",destination:"Bhubaneswar",ticket_cost:2500})
WriteResult({ "nInserted" : 1 })
```

65.
```
> db.bus.insertMany([{bus_id:452,name:"anusha",start_place:"Banglore",destination:"Chennai",ticket_cost:4500},{bus_id:32
1,name:"teja",start_place:"Raipur",destination:"Jodhpur",ticket_cost:4500}])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("62316e3b45c23ff7cbde1927"),
                ObjectId("62316e3b45c23ff7cbde1928")
        ]
}
> db.bus.find()
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230e211fc8fd25c07c5c47b"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("62316d6945c23ff7cbde1926"), "bus_id" : 452, "name" : "sita", "start_place" : "Vijayawada", "destinat
ion" : "Bhubaneswar", "ticket_cost" : 2500 }
{ "_id" : ObjectId("62316e3b45c23ff7cbde1927"), "bus_id" : 452, "name" : "anusha", "start_place" : "Banglore", "destinat
ion" : "Chennai", "ticket_cost" : 4500 }
{ "_id" : ObjectId("62316e3b45c23ff7cbde1928"), "bus_id" : 321, "name" : "teja", "start_place" : "Raipur", "destination"
 : "Jodhpur", "ticket_cost" : 4500 }
>
```

66.
```
> db.bus.update({"name":"sita"},{$set:{"ticket_cost":3000}})
WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1 })
```

```
> db.bus.updateMany({"ticket_cost":4500},{$set:{"ticket_cost":4575}})
{ "acknowledged" : true, "matchedCount" : 2, "modifiedCount" : 2 }
> db.bus.find()
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230e211fc8fd25c07c5c47b"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("62316d6945c23ff7cbde1926"), "bus_id" : 452, "name" : "sita", "start_place" : "Vijayawada", "destinat
ion" : "Bhubaneswar", "ticket_cost" : 3000 }
{ "_id" : ObjectId("62316e3b45c23ff7cbde1927"), "bus_id" : 452, "name" : "anusha", "start_place" : "Banglore", "destinat
ion" : "Chennai", "ticket_cost" : 4575 }
{ "_id" : ObjectId("62316e3b45c23ff7cbde1928"), "bus_id" : 321, "name" : "teja", "start_place" : "Raipur", "destination"
 : "Jodhpur", "ticket_cost" : 4575 }
>
```

67.

```
> db.bus.deleteOne({"bus_id":321})
{ "acknowledged" : true, "deletedCount" : 1 }
> db.bus.find()
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230e211fc8fd25c07c5c47b"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("62316d6945c23ff7cbde1926"), "bus_id" : 452, "name" : "sita", "start_place" : "Vijayawada", "destinat
ion" : "Bhubaneswar", "ticket_cost" : 3000 }
{ "_id" : ObjectId("62316e3b45c23ff7cbde1927"), "bus_id" : 452, "name" : "anusha", "start_place" : "Banglore", "destinat
ion" : "Chennai", "ticket_cost" : 4575 }
>
```

68.

```
> db.bus.deleteMany({"bus_id":452})
{ "acknowledged" : true, "deletedCount" : 2 }
> db.bus.find()
{ "_id" : ObjectId("6230c7acfbd872e0c3bc86d7"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230c7d5fbd872e0c3bc86d8"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86d9"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230c7e2fbd872e0c3bc86da"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86db"), "bus_id" : 210, "name" : "satya", "start_place" : "Hyderabad", "destinat
ion" : "Delhi", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230cb36fbd872e0c3bc86dc"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230e211fc8fd25c07c5c47b"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
>
```

69.

70.
```
> db.bus.deleteOne({"start_place":"Hyderabad"})
{ "acknowledged" : true, "deletedCount" : 1 }
> db.bus.find()
{ "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230af00fbd872e0c3bc86d2"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d3"), "bus_id" : 200, "name" : "meghana", "start_place" : "Guntur", "destinati
on" : "Banglore", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b062fbd872e0c3bc86d4"), "bus_id" : 350, "name" : "harsha", "start_place" : "Guntur", "destinatio
n" : "Munnar", "ticket_cost" : 4000 }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
>
```

71.
```
> db.bus.deleteMany({"start_place":"Guntur"})
{ "acknowledged" : true, "deletedCount" : 2 }
> db.bus.find()
{ "_id" : ObjectId("6230ae63fbd872e0c3bc86d1"), "bus_id" : 142, "name" : "sravani", "start_place" : "Rajahmundry", "dest
ination" : "Visakhapatnam", "ticket_cost" : 2000 }
{ "_id" : ObjectId("6230af00fbd872e0c3bc86d2"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
>
```

72.
```
> db.bus.remove({ticket_cost:2000})
WriteResult({ "nRemoved" : 1 })
> db.bus.find()
{ "_id" : ObjectId("6230af00fbd872e0c3bc86d2"), "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination
" : "Hyderabad", "ticket_cost" : 4000, "Destination" : "Araku" }
{ "_id" : ObjectId("6230b31ffbd872e0c3bc86d6"), "bus_id" : 415, "name" : "vahini", "start_place" : "Yanam", "destination
" : "Andaman", "ticket_cost" : 4000 }
>
```

73.
```
> db.bus.remove({})
WriteResult({ "nRemoved" : 2 })
> db.bus.find()
>
```

74.
```
> db.dropDatabase()
{ "ok" : 1 }
> show dbs
admin   0.000GB
config  0.000GB
data    0.000GB
local   0.000GB
mydb    0.000GB
>
```

75.
```
> db.bus.find({bus_id:{$in:[150,200]}},{_id:0})
{ "bus_id" : 200, "name" : "meghana", "start_place" : "Narasaraopeta", "destination" : "Banglore", "ticket_cost" : 5000
}
{ "bus_id" : 150, "name" : "roopa", "start_place" : "Nuzvid", "destination" : "Hyderabad", "ticket_cost" : 2500 }
{ "bus_id" : 150, "name" : "satya", "start_place" : "Ahmedabad", "destination" : "Ooty", "ticket_cost" : 5500 }
{ "bus_id" : 200, "name" : "Vahini", "start_place" : "Mandapeta", "destination" : "Vijayawada", "ticket_cost" : 4500 }
>
```

76.
```
> db.bus.aggregate([{$group:{_id:"$bus_id",num_cost:{$sum:"$ticket_cost"}}}])
{ "_id" : 350, "num_cost" : 8000 }
{ "_id" : 150, "num_cost" : 8000 }
{ "_id" : 200, "num_cost" : 9500 }
{ "_id" : 142, "num_cost" : 2000 }
>
```

77.
```
> db.bus.aggregate([{$group:{_id:"$bus_id",num_cost:{$avg:"$ticket_cost"}}}])
{ "_id" : 350, "num_cost" : 8000 }
{ "_id" : 150, "num_cost" : 4000 }
{ "_id" : 200, "num_cost" : 4750 }
{ "_id" : 142, "num_cost" : 2000 }
>
```

78.
```
db.bus.aggregate([{$group:{_id:"$bus_id",num_cost:{$min:"$ticket_cost"}}}])
"_id" : 350, "num_cost" : 8000 }
"_id" : 142, "num_cost" : 2000 }
"_id" : 150, "num_cost" : 2500 }
"_id" : 200, "num_cost" : 4500 }
```

79.
```
> db.bus.aggregate([{$group:{_id:"$bus_id",num_cost:{$max:"$ticket_cost"}}}])
{ "_id" : 350, "num_cost" : 8000 }
{ "_id" : 142, "num_cost" : 2000 }
{ "_id" : 150, "num_cost" : 5500 }
{ "_id" : 200, "num_cost" : 5000 }
>
```

80.
```
> db.bus.aggregate([{$group:{_id:"$bus_id",num_cost:{$first:"$ticket_cost"}}}])
{ "_id" : 350, "num_cost" : 8000 }
{ "_id" : 150, "num_cost" : 2500 }
{ "_id" : 200, "num_cost" : 5000 }
{ "_id" : 142, "num_cost" : 2000 }
>
```

81.
```
> db.bus.aggregate([{$group:{_id:"$bus_id",num_cost:{$last:"$ticket_cost"}}}])
{ "_id" : 200, "num_cost" : 4500 }
{ "_id" : 142, "num_cost" : 2000 }
{ "_id" : 150, "num_cost" : 5500 }
{ "_id" : 350, "num_cost" : 8000 }
>
```